Programming Problems

1. Sensor Stabilization

A sensor produces *n* readings a[0], a[1], ..., a[n-1]. The absolute differences between successive readings are guaranteed to be non-increasing.

Write a function

```
int firstStable(double a[], int n, double tolerance)
```

that returns the first index *i* such that |a[i+1] - a[i]| <= tolerance.

Return -1 if there is no such index. Call this function from `main` and print the returned value.

Input

The first line contains an integer n (n >= 2).
The second line contains n real numbers.
The third line contains the tolerance, a positive real number.

Output

Print the index returned by `firstStable`.

2. Days Elapsed in a Year

Write a function

```
int daysElapsed(int day, int month)
```

that returns the number of days elapsed in a non-leap year up to and including the given date. Return -1 if the day or month is invalid.

Call the function from `main` and print its return value.

Input

Two integers: day and month.

Output

Print the number of days elapsed, or -1 for an invalid date.

3. Machine Vibration - RMS Value

A vibration sensor records the velocity of a machine component at n instants. The readings may be positive or negative. A common measure of vibration magnitude is the root mean square (RMS):

```
RMS = sqrt((a[0]^2 + a[1]^2 + ... + a[n-1]^2) / n)
```

Write a function

```
double rms(double a[], int n)
```

that returns the RMS value of the array. Array indexing starts at 0. Call the function from `main`.

Input

The first line contains an integer n.
The second line contains n real numbers.

Output

Print the RMS value rounded to two decimal places.